

Department of Physics, FNSPE, CTU in Prague
02YELMA - Electricity and Magnetism
Midterm 2 (A - Solution)

Q1: By cylindrical symmetry, the magnetic field is purely azimuthal, $\vec{B}(s) = B_\phi(s) \hat{\phi}$. Using Ampère's law for a circular loop of radius s ,

$$B_\phi(s)(2\pi s) = \mu_0 I_{\text{enc}}$$

We first relate J_0 to the total current I :

$$I = \int_0^R J_0 \left(\frac{s}{R} \right) (2\pi s ds) \Rightarrow \boxed{J_0 = \frac{3I}{2\pi R^2}}$$

For $s < R$ (inside the conductor):

$$I_{\text{enc}}(s) = \int_0^s J_0 \left(\frac{s'}{R} \right) (2\pi s' ds') = \frac{2\pi J_0 s^3}{3R}$$
$$B_\phi(s)(2\pi s) = \mu_0 \frac{2\pi J_0 s^3}{3R}$$

$$\boxed{\vec{B}(s) = \frac{\mu_0 I}{2\pi R^3} s^2 \hat{\phi} \quad (s < R)}$$

For $s > R$ (outside the conductor):

$$B_\phi(s)(2\pi s) = \mu_0 I$$

$$\boxed{\vec{B}(s) = \frac{\mu_0 I}{2\pi s} \hat{\phi} \quad (s > R)}$$

Q2: Between the plates, the changing electric field produces a magnetic field. Using Ampère-Maxwell law for a circular loop of radius $s < R$,

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \epsilon_0 \frac{d\Phi_E}{dt}$$
$$B_\phi(s)(2\pi s) = \mu_0 \epsilon_0 \frac{d}{dt} (E\pi s^2)$$
$$\vec{B}(s) = \frac{\mu_0 \epsilon_0 s}{2} \frac{dE}{dt} \hat{\phi}$$
$$\vec{S} = \frac{1}{\mu_0} \vec{E} \times \vec{B} = -\frac{\epsilon_0 s}{2} E \frac{dE}{dt} \hat{s}$$

which points radially inward, indicating energy flows into the capacitor. Energy enters through the cylindrical side surface of area $dA = (2\pi R ds)$

$$P_{\text{in}} = \int \vec{S} \cdot d\vec{a} = \frac{\epsilon_0 R}{2} E \frac{dE}{dt} (2\pi R ds) = \epsilon_0 \pi R^2 ds E \frac{dE}{dt} = \frac{dU}{dt}$$
$$\frac{dU}{dt} = \epsilon_0 \pi R^2 ds E \frac{dE}{dt} \Rightarrow \boxed{U(t) = \frac{1}{2} \epsilon_0 \pi R^2 ds E(t)^2}$$

Q3: Since the wave propagates in free space, it travels in the $+\hat{z}$ direction. The corresponding electric field is perpendicular to both $\vec{B}$ and the direction of propagation:

$$\vec{E}(z, t) = E_0 \sin(kz - \omega t) \hat{x}, \quad E_0 = cB_0, \quad c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

(a) The instantaneous Poynting vector is

$$\vec{S} = \frac{1}{\mu_0} \vec{E} \times \vec{B}, \quad S = \frac{1}{\mu_0} E_0 B_0 \sin^2(kz - \omega t) = \frac{cB_0^2}{\mu_0} \sin^2(kz - \omega t)$$

$$\boxed{\langle S \rangle = \frac{cB_0^2}{2\mu_0} \text{ since } \langle \sin^2(\cdot) \rangle = \frac{1}{2}.}$$

The area of the circular surface is $A = \pi a^2$. Therefore the average power carried by the wave is $P = \langle S \rangle A$

$$\boxed{P = \frac{cB_0^2}{2\mu_0} \pi a^2}$$

(b) The transmitted electric-field amplitude is $E_{\text{trans}} = E_0 \cos \theta$ with $\theta = \frac{\pi}{4}$. Thus,

$$E_{\text{trans}} = E_0 \cos\left(\frac{\pi}{4}\right) = \frac{E_0}{\sqrt{2}}$$

Since $E_0 = cB_0$, the transmitted magnetic-field amplitude is also reduced by the same factor:

$$\boxed{B_{\text{trans}} = \frac{B_0}{\sqrt{2}}}$$

Q4: Since the cylinder is metallic, the charge resides on its outer surface. The surface charge density is $\sigma = \frac{Q}{2\pi r h}$ because the curved surface area is $A = 2\pi r h$ (neglecting the end caps). As the cylinder rotates, the surface charges move with tangential speed $v = \omega r$, thus the surface current density is

$$K = \sigma v = \sigma \omega r = \frac{Q}{2\pi r h} \omega r = \frac{Q\omega}{2\pi h}$$

If we treat the cylinder as stacked current loops, a thin strip of width dz carries current $dI = K dz$. Each strip forms a circular loop of area $A_{\text{loop}} = \pi r^2$ whose magnetic dipole moment is $d\mu = dI \cdot A_{\text{loop}}$. Therefore

$$d\mu = K dz \cdot \pi r^2$$

Integrating over the full height:

$$\mu = \int_0^h K \pi r^2 dz = K \pi r^2 h$$

Substituting K :

$$\mu = \frac{Q\omega}{2\pi h} \cdot \pi r^2 h$$

which gives

$$\boxed{\mu = \frac{1}{2} Q \omega r^2} \Rightarrow \boxed{\vec{\mu} = \frac{1}{2} Q \omega r^2 \hat{z}}$$